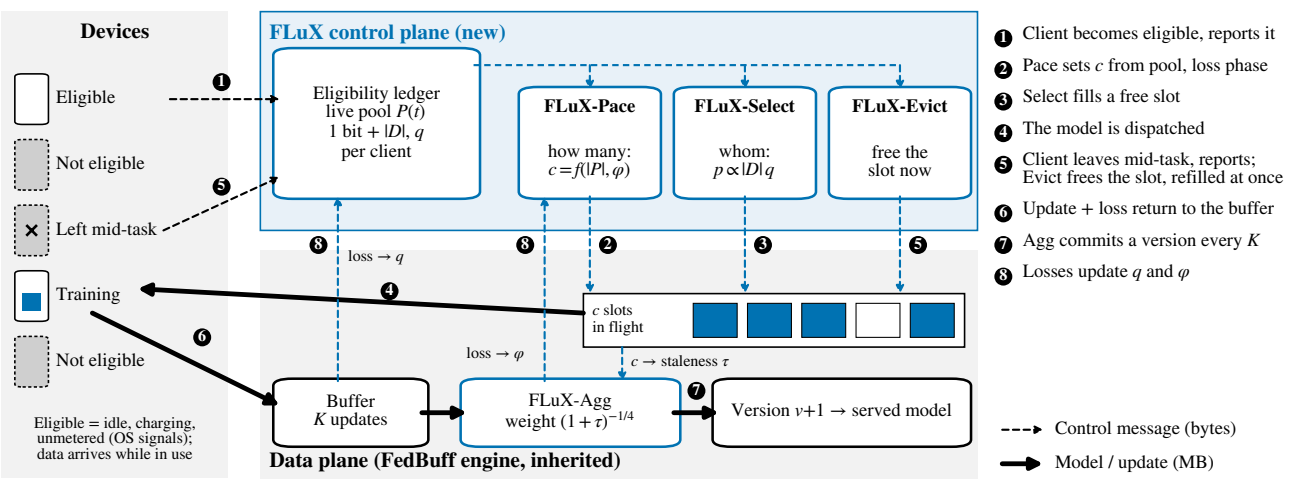
Devices
Eligible
Not eligible
Left mid-task
Training
Not eligible
Eligible = idle, charging, unmetered (OS signals); data arrives while in use
FLuX control plane (new)
Eligibility ledger
live pool $P(t)$
1 bit + $|D|$, q
per client
FLuX-Pace
how many:
$c = f(|P|, \varphi)$
FLuX-Select
whom:
$p \propto |D| q$
FLuX-Evict
free the
slot now
loss → q
loss → φ
c slots
in flight
c → staleness τ
Buffer
K updates
FLuX-Agg
weight $(1+\tau)^{-1/2}$
Version v+1 → served model
Data plane (FedBuff engine, inherited)
1 Client becomes eligible, reports it
2 Pace sets c from pool, loss phase
3 Select fills a free slot
4 The model is dispatched
5 Client leaves mid-task, reports; Evict frees the slot, refilled at once
6 Update + loss return to the buffer
7 Agg commits a version every K
8 Losses update q and φ
Control message (bytes)
Model / update (MB)